

Geomorphology & Oceanography Quick Stack

Plate tectonics, landforms by agents, currents, coral reefs and EEZ basics.

Portion focus ~40% · Topic: geo-physical

PYQ themes: Plate tectonics · Landforms · Ocean currents · Coral reefs

Why UPSC asks this

Diagrams-in-words and location–process matching recur in GS Prelims.

Solid earth & landforms

Interior layers, earthquakes/volcanoes at plate edges, weathering vs erosion, fluvial/karst/glacial/aeolian landforms. For India, know Himalayan vs Peninsular structural contrast.

Oceans

Currents (Gulf Stream, Humboldt, Labrador, Kuroshio — exemplar set), tides, EEZ (200 nm concept), coral reef conditions (temperature, salinity, clear water) and bleaching drivers.

Must remember

- Convergent/divergent/transform boundaries
- Himalayan orogeny = continental collision
- Warm vs cold currents impact climate/fisheries